

Degree: **B.Tech**, Stream: **CSE/CSE(I.O.T,CS & BCT)/CSIT/ECE/EEE/BT/CSE(A.I.M.L)/CST**
End Semester Examination 2023 – First Semester Examination, December-2023

Paper Code: BSCM103 Paper Name: Mathematics - I
Full Marks:100 Duration:3 Hours

Part - A
Attempt 10 questions
Each question carries 2 Marks (2 X 10)

1. Evaluate

2

$$\int_{-2}^2 |1 - x^2| dx.$$

or

1. Prove that

2

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

2. Show that

2

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = \frac{1}{6}.$$

or

2. Verify Rolle's Theorem for the function $f(x) = |x - 1|$ in the interval $[0, 2]$

2

3. If $u = f(r, \theta)$ where $x = r \cos \theta$, $y = r \sin \theta$, then show that $\frac{\partial u}{\partial x} = \cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta}$.

or

3. If $(\cos x)^y = (\sin y)^x$ find $\frac{dy}{dx}$.

4. State the Euler's theorem on homogeneous functions, of two variables.

or

4. What is the greatest rate of increase of $\phi = xyz^2$ at the point $(1, 0, 3)$?
5. If 3 and 15 are the two eigenvalues of the matrix

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

then find the third eigenvalue of the matrix.

or

5. Find for what values of

$$x, \begin{vmatrix} x + \omega^2 & \omega & 1 \\ \omega & 1 + x & \omega^2 \\ 1 & \omega^2 & x + \omega \end{vmatrix} = 0,$$

where ω stands for cube root of unity.

6. State Raabe's Test.

or

6. Show that the sequence is convergent.

$$\left\{ \frac{(-1)^n}{n} \right\}$$

7. Evaluate

2

$$\int_0^{\infty} e^{-x^2} x^9 dx.$$

or

7. Find for what values of x the function

2

$$f(x) = x^3 - 3x^2 + 6x + 3 \text{ has maximum or minimum.}$$

8. Using the concept of improper integrals, establish the relation

2

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi$$

or

8. Derive Lagrange's Mean Value Theorem from Cauchy's Mean Value Theorem.

2

9. Test if the function

2

$$f(x, y) = \left(\sqrt{xy} + y^{\frac{1}{4}} \right)^3$$

is a homogeneous function in x and y . If yes, find the degree of homogeneity.

or

9. If $u = e^{xyz}$ find $\frac{\partial^3 u}{\partial x \partial y \partial z}$ at $(1, 1, 1)$

2

10. Find the critical points of the function $f(x, y) = x^3 + y^3 - 3axy$.

2

or

10. When do we call a critical point maxima, minima or a saddle point?

2

Part - B
Attempt 6 questions
Each question carries 5 Marks (5 X 6)

11. If $u = f(x^2 + 2yz, y^2 + 2zx)$, prove
that $(y^2 - zx) \frac{\partial u}{\partial x} + (x^2 - yz) \frac{\partial u}{\partial y} + (z^2 - xy) \frac{\partial u}{\partial z} = 0.$

or

11. Find the directional derivative of the function $\phi = x^2 - y^2 + 2z^2$ at the point P (1,2,3) in the direction of the line PQ where Q is the point (5,0,4).
12. Find whether the following homogeneous system of equations has non-trivial solution and find them if possible:

$$\begin{aligned}x + 2y + 3z &= 0, \\2x + 3y + z &= 0, \\3x + y &= -2z.\end{aligned}$$

or

12. Verify the Cayley-Hamilton theorem for the matrix

$$\begin{bmatrix} -2 & 4 \\ 4 & 4 \end{bmatrix}$$

13. Prove that the sequence $\{n^{\frac{1}{n}}\}$ is bounded.

or

13. Is the series convergent? Justify.

$$\sum_{n=1}^{\infty} \sin \frac{1}{n}$$

14. Verify Euler's theorem on homogeneous function for:

$$z = \frac{x^{\frac{1}{4}} + y^{\frac{1}{4}}}{x^{\frac{1}{5}} + y^{\frac{1}{5}}}.$$

or

14. If $u = \tan^{-1} \frac{y}{x}$, where $x = e^t - e^{-t}$ and $y = e^t + e^{-t}$, find $\frac{du}{dt}$. 5

15. Find the eigenvalues and eigenvectors of the matrix 5

$$\begin{bmatrix} 4 & 3 \\ 2 & 9 \end{bmatrix}$$

or

15. Prove that 5

$$\begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 1+b & 1 & 1 \\ 1 & 1 & 1+c & 1 \\ 1 & 1 & 1 & 1+d \end{vmatrix} = abcd \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right)$$

16. Examine whether the following series is convergent or not: 5

$$\left(\frac{1}{3} \right)^2 + \left(\frac{1 \times 2}{3 \times 5} \right)^2 + \left(\frac{1 \times 2 \times 3}{3 \times 5 \times 7} \right)^2 + \dots \infty$$

or

16. Test the convergence of the series 5

$$\frac{1}{4 \times 7 \times 10} + \frac{4}{7 \times 10 \times 13} + \frac{9}{10 \times 13 \times 16} + \dots \infty$$

Part - C

Attempt 5 questions

Each question carries 10 Marks (10 X 5)

17. Examine whether the integral 10

$$\int_{-1}^{+1} \frac{dx}{x^2},$$

exists or not in the general sense. Does it exist in the sense of Cauchy's Principal value? Justify your answer.

OR

17. Find the surface area and volume of the ellipsoid formed by the revolution of

10

the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ about its major axis.

18. If $x = r \cos \theta$ and $y = r \sin \theta$, prove that

10

a. $\left(\frac{\partial r}{\partial x}\right)^2 + \left(\frac{\partial r}{\partial y}\right)^2 = 1.$

b. $\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} = \frac{1}{r}.$

c. $\frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} = 0.$

OR

18. a. Determine whether the following limit exists or not:

10

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^4}{x^2 - y^4}.$$

b. Verify whether the following function is continuous at (0,0):

$$f(x, y) = \begin{cases} x \sin\left(\frac{1}{y}\right) + y \sin\left(\frac{1}{x}\right), & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0). \end{cases}$$

19. Find the eigenvalues and corresponding eigenvectors of the following matrix:

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$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

or

19. Use Gauss-Jordan method to obtain the inverse of the following matrix:

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$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

20.

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Show that the sequence $\{\sqrt{7}, \sqrt{7 + \sqrt{7}}, \sqrt{7 + \sqrt{7 + \sqrt{7}}}, \dots, \infty\}$ is monotonically increasing and bounded above. Also find the limit of the sequence.

or

20. Show that the sequence

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$\{x_n\}$ defined by $x_n = \frac{3n+1}{n+2}$ is strictly increasing. Is it bounded?

Comment on the convergence of the sequence with proper justification. If the limit exists, then find the limit.

21. a) Express in terms of gamma function:

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$$\int_0^1 \frac{dx}{\sqrt{1-x^4}}$$

b) Evaluate, if possible, the improper integral $\int_{-1}^1 \sqrt{\frac{1+x}{1-x}} dx$

or

21. Expand $\tan x$ by Maclaurin's series upto and including the term containing x^5 .
By choosing $x=0.3$ radians, compare the value generated by this series with the true value as given by the scientific calculator. Evaluate the error correct to four decimal places.
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10